# The Riemann Kernel Is Globally $\mathrm{PF}_4$

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**Abstract**

Let $\Phi$ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that its exact global Pólya-frequency order is four: every translation minor of order at most four is strictly positive at strictly ordered nodes, whereas the order-five minor at the exact spacing 211/2000 is negative. Both halves of this classification are exported by a maintained Lean 4 theorem; the proof depends only on the standard Lean/mathlib axioms. The optional order-five confluent coefficient changes sign exactly once on the positive half-line, at $0.0622795266356 < u_* < 0.0622795266357$. For the positive half, a two-mode argument proves global positivity of the log-curvature quantities $q, F_2$ and of the fully confluent invariant $\mathcal{C}_4$. The finite verification uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The direct finite-witness certificate likewise uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w \Delta(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} \, \mathrm{d}t.$$

Here $\Delta$ is a deterministic closed cumulative gap built from two explicitly normalized, curvature-tilted triangular weights. Its strict positivity follows on the actual coordinate interval from the exact mass identities and one explicit algebraic sign change; no probability or CDF premise is required. The transport reductions and analytic sign estimates are given explicitly, with a table of the finite polynomial inequalities used for the three global signs. We then construct an explicit positive even Schwartz kernel which is itself strictly $\mathrm{PF}_4$ but whose entire Fourier transform has nonreal zeros; the latter conclusion follows from a short exact discriminant inequality. This delimits the finite-order reach of Schoenberg's infinite-order transform correspondence.

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# 1 Introduction

A function $f : \mathbb{R} \to \mathbb{R}$ is $\mathrm{PF}_r$ when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^{k}$$

is nonnegative for $1 \le k \le r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Pólya-frequency order of the Riemann kernel; see [3] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

**Theorem 1.1** (Exact global PF order)**.** *The exact global Pólya-frequency order of $\Phi$ is four. More precisely:*

1. *for every* $1 \le k \le 4$ *and strictly increasing real tuples* $x_1 < \cdots < x_k$ *and* $y_1 < \cdots < y_k$*,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^k > 0.$$

2. *with* $h = 211/2000$*,*

$$\det[\Phi((i-j)h)]_{i,j=0}^4 < 0.$$

The two theorem inputs are deliberately separated. The positive input is the universal strict-order theorem through four. The negative input is the direct rational order-five witness in part (2), including proved strict ordering of its nodes and the signed Toeplitz-to-translation-matrix identification. In the maintained formal development these are assembled by `PF4.globalRiemannKernel_pfOrderExactly_four`; its axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`.

The exact-order classification and the Fourier question are distinct. The following example marks a precise finite-order boundary for Schoenberg's infinite-order transform correspondence, even within the positive even Schwartz class.

**Theorem 1.2** (Continuous $\mathrm{PF}_4$ Fourier separator)**.** *There is an explicit positive even Schwartz function* $f_*$ *which is strict* $\mathrm{PF}_4$*, while its entire Fourier transform has nonreal zeros. Consequently continuous* $\mathrm{PF}_3$ *and continuous* $\mathrm{PF}_4$ *membership are each insufficient to force Fourier real-rootedness.*

The proof has two one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict $\mathrm{PF}_3$ and positivity of a three-point functional $\Lambda$. Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict $\mathrm{PF}_4$ to the monotonicity of a scalar three-point function $\Psi$. On the actual image of an increasing curvature coordinate, the derivative of $\Psi$ becomes a positive integral of $\mathcal{C}_4$ against a deterministic closed cumulative gap.

The finite part of the positive input consists of exact rational Bernstein coefficients on fixed algebraic domains, followed by analytic bounds for all remaining theta modes; Appendix B lists the transformed target polynomials, domains, degrees, minimum coefficients, derivative envelopes, and the surviving relative margins. The transport algebra is proved in the text, and the order-five calculation uses rational enclosures and elementary series bounds. An optional recurrence gives a global order-five refinement: its confluent coefficient has one positive zero, enclosed in a rational interval of width $10^{-13}$, and an explicit rational spacing gives a comfortably negative finite determinant. Neither refinement is needed for the strict-$\mathrm{PF}_4$ theorem; it records additional structure around the direct witness. Theorem 1.2, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

## 2 Relation to earlier work

Schoenberg introduced Pólya-frequency functions and connected translation minors with bilateral Laplace transforms and variation-diminishing convolution operators [1, 3]. Karlin's monograph remains the standard systematic account of total positivity and its composition principles [2]; a modern review of PF functions and their preservers is given by Belton, Guillot, Khare, and Putinar [3].

Finite-order translation kernels have a substantially wider range than the infinite-order class. Khare's characterization of measurable $\mathrm{TN}_p$ functions for $p \ge 3$ provides a general finite-order setting [4]. The present proof instead derives a criterion specific to order four, using successive quotient derivatives and a curvature transport identity.

The theta normalization used here is the classical one in de Bruijn's study of the Riemann Fourier kernel [5]. For the same kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [6], while Csordas and Varga obtain strict concavity and moment inequalities connected with the Riemann Hypothesis [7]. Grochenig explains the distinct Schoenberg correspondence relevant to the Riemann Hypothesis: the PF function there is recovered from the reciprocal of a Laguerre–Pólya entire function, rather than identified with the original Riemann kernel [8].

Michałowski first published an explicit negative $5 \times 5$ Toeplitz minor for this kernel, supported by an interval computation, together with single-configuration positivity checks at orders at most four [9]. Global $\mathrm{PF}_4$ and a symbolic proof of the order-five leading sign are posed there as Open Problems 1 and 2. Theorem 1.1 resolves the first, and Lemma 11.2 resolves the second in strengthened form. Indeed, for the Toeplitz determinant

$$D_r(u_0, h) = \det[\Phi(u_0 + (i-j)h)]_{i,j=0}^{r-1},$$

Lemma 6.2 gives its leading coefficient as

$$C_r(u_0) = \frac{(-1)^{r(r-1)/2} V(0, \dots, r-1)^2}{\prod_{k=0}^{r-1} (k!)^2} \det[\Phi^{(i+j)}(u_0)]_{i,j=0}^{r-1}.$$

For $r = 5$ the sign factor is positive, only derivatives through order eight occur, and the limit as $u_0 \to 0^+$ is a positive multiple of the parity-factorized determinant $\mathsf{H}_5 = CB < 0$ computed exactly in Lemma 11.2. Theorem 11.3 determines the complete sign change: $C_5(u_0) < 0$ precisely for $0 \le u_0 < u_*$, where $0.0622795266356 < u_* < 0.0622795266357$. The normalization in [9] uses $K_M(u) = \Phi(2u)/2$, so its corresponding center is $u_*/2 \in (0.0311397633178, 0.03113976331785)$.

Theorem 1.2 further shows where the infinite-order transform conclusion ceases to follow from finite-order positivity: even strict $\mathrm{PF}_3$ or $\mathrm{PF}_4$ does not control all Fourier zeros.

## 3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \qquad \Theta(t) = \vartheta(e^{2t}), \qquad H(t) = e^{t/2}\Theta(t).$$

This is a real-valued definition: summability is first proved directly for every $x > 0$, before any analytic continuation is invoked. The real specialization of Gaussian Poisson summation gives the exact normalization $\vartheta(x) = x^{-1/2}\vartheta(1/x)$, and hence

$$\Theta(-t) = e^t \Theta(t), \qquad H(-t) = H(t).$$

A Jacobi-theta realization supplies a globally analytic function agreeing with this real series. Thus $H$ is real analytic and even. Define

$$\Phi(t) = \frac{1}{2}\left(H''(t) - \frac{1}{4}H(t)\right).$$

This is twice the common de Bruijn kernel normalization; multiplication by a positive constant changes none of the PF signs or strictness statements. On $t \ge 0$, locally uniform polynomial–geometric majorants justify termwise differentiation through order six and give

$$\Phi(t) = \sum_{n \ge 1} \left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2}\right) e^{-\pi n^2 e^{2t}}. \tag{3.1}$$

Parity reflects this identity to every real $t$, equivalently by replacing $t$ on the right by $|t|$. Thus the positive-side formula is the restriction of a real-analytic even function. Smoothness and every derivative used later at the origin are consequences of that analytic realization, not of numerical coverage or of differentiating an absolute-value definition.

For $t \geq 0$, the coefficient of the $n$-th exponential can be factored as

$$2\pi n^2 e^{5t/2}\bigl(2\pi n^2 e^{2t} - 3\bigr) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \qquad s = \ell', \qquad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a,b) = s(a) - s(b), \qquad M(a,b) = \frac{q(b) - q(a)}{A(a,b)}. \tag{3.2}$$

When $q > 0$, $s$ is strictly decreasing and $A(a,b) > 0$.

The maintained formal boundary mirrors this order of construction. It separately checks the Poisson step, analytic continuation, parity, reflected series, and the complete jet through order six. The exact declarations are listed in Appendix D; none uses a project-specific axiom.

# 4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned}
&m_0 = 1, \quad m_1 = 0, \quad m_2 = c_2, \quad m_3 = c_3,\\
&m_4 = c_4 + 3c_2^2, \quad m_5 = c_5 + 10c_2c_3,\\
&m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3,
\end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \tag{4.1}$$

**Lemma 4.1** (Certified input)**.** *For the kernel* $\Phi$*,*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof combines exact rational polynomial inequalities with analytic theta-tail bounds. For $u \geq 0$, put

$$x = \pi e^{2u}, \qquad \eta = e^{-3x}, \qquad R_0(z) = 2z - 3,$$

and recursively

$$R_{j+1}(z) = \left(\frac{5}{2} - 2z\right) R_j(z) + 2zR_j'(z).$$

After division by the positive first theta mode, the first two modes give the normalized jet entries

$$a_j(x,\eta) = \frac{R_j(x) + 4\eta R_j(4x)}{R_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000.$$

Here the half-line certificates are applied directly to the cleared numerators of the shifted targets $q_{1,2} - 10$, $(F_2)_{1,2} - 1000$, and $(\mathcal{C}_4)_{1,2} - 50{,}000{,}000$. The complete contribution of modes $n \geq 3$ is bounded by one monotone polynomial-geometric estimate. Its changes in the same cleared sign numerators are smaller than the three displayed margins, both on $\pi \leq x \leq 5$ and on $x \geq 5$. Evenness completes the real line. All inequalities, including the derivative envelopes used on the tail, are derived in Appendix B. The exact replay uses no floating point, no interval subdivision, and no FLINT or Arb arithmetic.

In the maintained Lean development the compact and outer regions are one universal theorem, rather than sampled checks. The three positive-sign declarations prove the cleared $q, F_2, \mathcal{C}_4$ numerators for every $t \geq 0$. Exact raw-jet identities then remove the positive denominators, and the reflected global jet from Section 3 covers the negative half-line and the origin. The Python verifier remains an independent exact replay of the printed rational certificate; it is not the bridge from sampled arithmetic to the universal statement.

# 5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{5.1}$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a, b) = \frac{\int_a^b q'(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t} = \frac{\int_a^b f_0(t)q(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t};$$

thus $M(a, b)$ is a $q$-weighted mean of $f_0$. Choose a point $u \in [\xi, m]$ where $f_0$ is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. The weighted-mean bounds and the ordering of the extremizers are

$$M(\xi, m) \geq f_0(u), \qquad M(m, r) \leq f_0(v), \qquad u \leq m \leq v.$$

Consequently

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_\xi^r \max(f_0'(t), 0)\,\mathrm{d}t.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_\xi^r q(t)\,\mathrm{d}t$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_\xi^r \left( q - \frac{\max(F_1, 0)}{q^2} \right) \mathrm{d}t \tag{5.2}$$

$$= \int_\xi^r \frac{q^3 - \max(F_1, 0)}{q^2}\,\mathrm{d}t = \int_\xi^r \frac{\min(q^3, F_2)}{q^2}\,\mathrm{d}t > 0. \tag{5.3}$$

The second equality uses $q^3 - \max(F_1, 0) = \min(q^3, q^3 - F_1) = \min(q^3, F_2)$; the last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global $\mathrm{PF}_3$. We rederive the needed Wronskian form in the next section, where the quotient integral transfers these confluent signs to every ordered collocation minor.

# 6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \qquad w_j = (v_j/v_2)'.$$

These quotients are introduced only after their denominators are known to be positive: $u_1 > 0$ by Section 3, and $v_2 > 0$ by the next display. Later division by $w_3$ is likewise made only after (6.8) proves $w_3 > 0$. Strict log-concavity gives

$$v_j = \frac{u_j}{u_1} A(p_j, p_1) > 0 \qquad (j \geq 2).$$

The case $k = 2$ of Lemma 6.1 below therefore proves strict order two directly. At the fixed orders used in this paper, successive column division and differentiation give the exact three-stage cascade

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{6.1}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 W(v_2, v_3, v_4), \tag{6.2}$$

$$W(v_2, v_3, v_4) = v_2^3 W(w_3, w_4), \tag{6.3}$$

$$W(w_3, w_4) = w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.4}$$

Consequently

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.5}$$

No sign premise is used in these algebraic identities beyond the stated nonvanishing needed to form the quotients.

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T} A(a, b) = q(b) - q(a), \qquad \mathcal{T} \log A(a, b) = M(a, b). \tag{6.6}$$

The sign convention in (6.6) will be used below. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \tag{6.7}$$

The exact weighted-mean split

$$A(p_j, p_1) M(p_j, p_1) = A(p_j, p_2) M(p_j, p_2) + A(p_2, p_1) M(p_2, p_1)$$

turns (6.7) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \tag{6.8}$$

Thus $w_j = (v_j/v_2)g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \tag{6.9}$$

Equation (6.8) and (6.6) give

$$\mathcal{T} \log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T} \log \Lambda_j,$$

where $\Lambda_j = \Lambda(p_j; p_2, p_1)$. A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (6.9), yielding the exact identity

$$\mathcal{T} \log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \tag{6.10}$$

**Lemma 6.1** (Iterated quotient integral)**.** *Let $f_1, \dots, f_k$ be continuously differentiable on an interval, let $f_1 > 0$, and put $G_j = f_j/f_1$. For $t_1 < \cdots < t_k$,*

$$\det[f_j(t_i)]_{i,j=1}^{k} = \prod_{i=1}^{k} f_1(t_i) \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[G'_{j+1}(\tau_i)]_{i,j=1}^{k-1} \, \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{k-1}. \tag{6.11}$$

*The identity may be applied repeatedly whenever the first member of the new derivative family is positive.*

*Proof.* Factor $f_1(t_i)$ from row $i$. The first column is then one. Replace successive rows, from bottom to top, by their difference with the preceding row. Each remaining entry is $G_j(t_{i+1}) - G_j(t_i) = \int_{t_i}^{t_{i+1}} G'_j(\tau) \, \mathrm{d}\tau$. Determinant multilinearity gives (6.11). Since $\tau_i \in [t_i, t_{i+1}]$, the integration nodes remain ordered, so the same argument applies to the derivative determinant. $\square$

Only the instances of Lemma 6.1 of sizes two through four enter the theorem. The maintained formal engine proves exactly those instances and the corresponding strict integral boxes; no arbitrary-order quotient theorem is a premise of Theorem 1.1. In particular, `PF4.FixedOrderQuotientWronskian.fixedOrderFour_quotientWronskian_package` packages (6.1)–(6.4) together with the reversed endpoint orientation $p_4 < p_3 < p_2 < p_1$. Its independent replay boundary is `CERT29`.

**Lemma 6.2** (Confluent divided-difference limit)**.** *Let $f \in C^{2k-2}$, let $V(a) = \prod_{0 \le i < j < k}(a_j - a_i)$, and let $a, b$ be fixed strictly increasing $k$-tuples. Then*

$$\lim_{h \downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)} V(a) V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1} (r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}. \tag{6.12}$$

*If only the row nodes coalesce while $y_0 < \cdots < y_{k-1}$ remain fixed, then*

$$\lim_{h \downarrow 0} \frac{\det[f(z + h a_i - y_j)]_{i,j=0}^{k-1}}{h^{k(k-1)/2} V(a)} = \frac{\det[f^{(i)}(z - y_j)]_{i,j=0}^{k-1}}{\prod_{r=0}^{k-1} r!}. \tag{6.13}$$

*Proof.* Apply successive Newton divided-difference row operations, whose divisors are the positive numbers $h(a_j - a_i)$, and let $h \downarrow 0$. This gives (6.13). Applying the same operation to the columns gives derivatives with respect to $-y$, hence the factor $(-1)^{0+\cdots+(k-1)}$, and proves (6.12). This records the factorial normalization and does not assume a nonzero leading coefficient. ☐

**Proposition 6.3** (Three-point equivalence)**.** *Under the strict lower-order positivity already proved, global* $\mathrm{PF}_4$ *is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \le 0 \qquad (\xi < m < r).$$

*Strict inequality implies strict global* $\mathrm{PF}_4$.

*Proof.* If $\partial_\xi \Psi \le 0$, then $p_4 < p_3$ makes the right side of (6.10) nonnegative. All prefactors in (6.5) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For clarity, applying Lemma 6.1 three times to the order-four minor gives, with $I_i = [t_i, t_{i+1}]$,

$$\det[u_j(t_i)]_{i,j=1}^4 = \prod_{i=1}^4 u_1(t_i) \int_{I_1 \times I_2 \times I_3} \prod_{i=1}^3 v_2(\tau_i)$$
$$\times \int_{\tau_1}^{\tau_2} \int_{\tau_2}^{\tau_3} w_3(\sigma_1) w_3(\sigma_2) \int_{\sigma_1}^{\sigma_2} \left(\frac{w_4}{w_3}\right)'(\rho)\, \mathrm{d}\rho\, \mathrm{d}\sigma_2\, \mathrm{d}\sigma_1\, \mathrm{d}\boldsymbol{\tau}.$$

Every factor is positive in the strict case, and every integration interval has positive length. This proves positivity of every ordered collocation minor, not only of the confluent Wronskian.

Conversely, assume global $\mathrm{PF}_4$. Fix arbitrary $\xi_1 < \xi_2 < m < r$, choose any $t$, and choose the columns so that $(p_4, p_3, p_2, p_1) = (\xi_1, \xi_2, m, r)$ at $t$; explicitly, $y_j = t - p_j$, which preserves $y_1 < \cdots < y_4$. Coalesce arbitrary strictly ordered row nodes at $t$. Equation (6.13) and global $\mathrm{PF}_4$ make the resulting translate Wronskian nonnegative. Equations (6.5) and (6.10) therefore give

$$\Psi(\xi_1; m, r) \ge \Psi(\xi_2; m, r).$$

Because both the pair $\xi_1 < \xi_2$ and the pair $m < r$ were arbitrary, $\Psi(\cdot; m, r)$ is nonincreasing for every admissible $m, r$. Smoothness makes this equivalent to the stated differential inequality. ☐

# 7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \qquad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $\mathrm{d}y/\mathrm{d}t = q(t) = Q(y) > 0$. Primes in this and the next three sections mean $y$-derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \qquad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a, b) = y(b) - y(a)$ and $M(a, b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \tag{7.1}$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L}\int_p^z (y-p)\kappa(y)\,\mathrm{d}y + \frac{1}{R}\int_z^w (w-y)\kappa(y)\,\mathrm{d}y, \tag{7.2}$$

$$\delta := -\partial_p\Lambda = \frac{1}{L^2}\int_p^z (z-y)\kappa(y)\,\mathrm{d}y > 0. \tag{7.3}$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w,$$

and $\partial_\xi = Q(p)\partial_p$. Thus $\Lambda_\xi = -Q(p)\delta$.

Define

$$\mathcal{N} = \delta\Lambda^2 + Q'(p)\delta\Lambda + \Lambda\mathcal{T}\delta - \delta\mathcal{T}\Lambda. \tag{7.4}$$

Differentiating $\Psi = \Lambda + \mathcal{T}\log\Lambda$, commuting simultaneous translation with $\partial_\xi$, and collecting over $\Lambda^2$ gives

$$\partial_\xi\Psi = -\frac{Q(p)}{\Lambda^2}\mathcal{N}. \tag{7.5}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{7.6}$$

It remains to prove $K > 0$.

All objects here are defined only on the image of $y = -s$. The range-sign theorem in `PF4.PaperObjectClosure` proves the displayed $\rho, \kappa$ inequalities at every actual coordinate point. The package in `PF4.SimultaneousCoordinateTranslation` identifies literal simultaneous translation of the original endpoints with $\mathcal{T}$, including the derivative of the nonconstant coefficient $Q(p)$. Neither result extends $Q$ by an assumed inverse or asserts that the coordinate image is all of $\mathbb{R}$.

# 8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log\kappa)'\}'. \tag{8.1}$$

**Lemma 8.1** (Confluent-to-curvature identity)**.** *The determinant* (4.1) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{8.2}$$

*Consequently $D > 0$ on $\mathbb{R}$.*

*Proof.* The change of variable gives $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out $Q^6$. The remaining factor is

$$\kappa^2\left(3(\kappa-1) - \{Q(\log\kappa)'\}'\right),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. □

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by $\Phi^4$. The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

# 9 Exact endpoint transport identity

Write the three normalized triangular densities from (7.2) and (7.3) as

$$\mu_L(y) = \frac{(z-y)\kappa(y)}{L^2\delta} \quad (p \le y \le z), \tag{9.1}$$

$$\nu_L(y) = \frac{(y-p)\kappa(y)}{L\Lambda} \quad (p \le y \le z),$$

$$\nu_R(y) = \frac{(w-y)\kappa(y)}{R\Lambda} \quad (z \le y \le w). \tag{9.2}$$

The triangular identities give

$$\int_p^z \mu_L = 1, \qquad \int_p^z \nu_L + \int_z^w \nu_R = 1. \tag{9.3}$$

No measure theory is needed below; these are normalized continuous weights on compact intervals.

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y)\frac{\kappa'(y)}{\kappa(y)}.$$

Then $A_0' = D$.

**Lemma 9.1** (Endpoint transport cancellation)**.** *With $K$ as in* (7.6)*, put*

$$\mathcal{E} = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{L\mathcal{H}(p) - I_L}{L^2\delta}, \tag{9.4}$$

*where $I_L = \mathcal{J}(z) - \mathcal{J}(p)$ and $I_R = \mathcal{J}(w) - \mathcal{J}(z)$, with $\mathcal{H}, \mathcal{J}$ defined below. Then*

$$K = \mathcal{E}. \tag{9.5}$$

*Proof.* Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = \mathcal{H}', \qquad \mathcal{H} = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \tag{9.6}$$

$$\mathcal{H} = \mathcal{J}', \qquad \mathcal{J} = y^3 - 3yQ + QQ'. \tag{9.7}$$

Both identities follow by one differentiation. Integrating the three linear weights once by parts gives

$$\int_p^z A_0\nu_L + \int_z^w A_0\nu_R - \int_p^z A_0\mu_L = \mathcal{E}. \tag{9.8}$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \quad \mathcal{T}\delta = \frac{U(p,z) - (Q(z)-Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}. \tag{9.9}$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{9.10}$$

The endpoint identity equating (9.4) and (9.10), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of $Q, Q', Q''$; no bound, coordinate-surjectivity assertion, or polynomial model is used. □

The normalized weights may independently be regarded as probability measures $\mu$ and $\nu$. In that notation (9.8) says $\mathcal{E} = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]$. This interpretation is formally checked, but the deterministic proof in the next section uses only the displayed compact-interval densities and their masses.

## 10 The positive closed coordinate gap

Define a deterministic piecewise function on $[p, w]$ by

$$\Delta(y) = \begin{cases} \dfrac{-(z-y)^2 - (z-y)Q'(y) - Q(y) + (z-p)^2 + (z-p)Q'(p) + Q(p)}{L^2\delta} \\ \qquad - \dfrac{(y-p)^2 - (y-p)Q'(y) + Q(y) - Q(p)}{L\Lambda}, & p \le y \le z, \\ \dfrac{(w-y)^2 + (w-y)Q'(y) + Q(y) - Q(w)}{R\Lambda}, & z < y \le w. \end{cases} \tag{10.1}$$

The fundamental theorem of calculus and $\kappa = 2 - Q''$ identify these closed expressions with

$$\Delta(y) = \begin{cases} \displaystyle\int_p^y (\mu_L(t) - \nu_L(t))\,\mathrm{d}t, & p \le y \le z, \\ \displaystyle\int_y^w \nu_R(t)\,\mathrm{d}t, & z < y \le w. \end{cases} \tag{10.2}$$

The two branches agree at $z$ by (9.3).

**Lemma 10.1** (Strict closed gap)**.** *The function $\Delta$ is continuous on $[p, w]$, satisfies $\Delta(p) = \Delta(w) = 0$, and*

$$\Delta(y) > 0 \qquad (p < y < w).$$

*Proof.* The endpoints and continuity follow from (10.1), (10.2), and the exact mass identities. On $(p, z)$, the difference of the positive left densities changes sign at the explicit strict convex combination

$$y_* = \frac{\Lambda z + L\delta p}{\Lambda + L\delta} \in (p, z).$$

Indeed, clearing the positive common factor $\kappa(y)$ and the positive denominators gives

$$\mu_L(y) - \nu_L(y) > 0 \iff y < y_*, \qquad \mu_L(y) - \nu_L(y) < 0 \iff y > y_*.$$

Thus $\Delta(y) > 0$ for $p < y \le y_*$. For $y_* \le y \le z$, subtract the two identities in (9.3) and split at $y$ to obtain

$$\Delta(y) = \int_y^z (\nu_L - \mu_L)\,\mathrm{d}t + \int_z^w \nu_R\,\mathrm{d}t > 0.$$

The first integral is nonnegative and the second is strictly positive because $z < w$, $\kappa > 0$, and $\Lambda > 0$. On $(z, w)$, positivity follows directly from the positive tail integral in (10.2). $\square$

On the left branch $\Delta' = \mu_L - \nu_L$, and on the right branch $\Delta' = -\nu_R$. Direct integration by parts on $[p, z]$ and $[z, w]$, using $A_0' = D$, the zero endpoint values, and the equality of the two branches at $z$, gives

$$\begin{aligned} \int_p^w \Delta(y) D(y)\,\mathrm{d}y &= \int_p^z A_0 \nu_L + \int_z^w A_0 \nu_R - \int_p^z A_0 \mu_L \\ &= \mathcal{E} = K. \end{aligned} \tag{10.3}$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w \Delta(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2} \,\mathrm{d}y > 0. \tag{10.4}$$

Every factor in the interior is strictly positive.

The probability interpretation is optional: the same $\Delta$ is $F_\mu - F_\nu$, and the density ratio has one crossing. The proof above does not require probability instances, CDF complement identities, endpoint ratio limits, or a separate existence-and-uniqueness argument for that crossing. The two actual-kernel exports in `PF4.GlobalStrictPF4` prove, for every $\xi < m < r$, respectively the strict positivity in (10.4) and the negative derivative in (7.5). All coordinate points in these statements lie in the actual image of $y = -s$; no surjectivity onto $\mathbb{R}$ is asserted.

# 11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 6.3 therefore proves part (1) of Theorem 1.1.

We now prove directly that the classification stops at order four. Let $h = 211/2000$. Exact rational theta enclosures give

$$-1.08 \cdot 10^{-7} < \det[\Phi((i-j)h)]_{i,j=0}^{4} < -1.05 \cdot 10^{-7}. \tag{11.1}$$

**Theorem 11.1.** *The Riemann kernel is not* $\mathrm{PF}_5$.

*Proof.* The nodes $0, h, 2h, 3h, 4h$ are strictly increasing. Evenness identifies the translation matrix with the symmetric Toeplitz matrix determined by the five kernel values $\Phi(0), \Phi(h), \ldots, \Phi(4h)$. Each value is enclosed by exact Taylor bounds for the retained theta modes and the maintained infinite tail theorem; a rational determinant parity factorization then gives (11.1). This is a negative translation minor at distinct nodes, with no limiting argument.

In the formal development the node ordering, signed-index orientation, Toeplitz identification, value boxes, determinant sign, and primary-kernel identity are all premises of `PF4.globalRiemannKernel_orderFive_translationMinor_neg`. Thus the order-five obstruction used by Theorem 1.1 does not depend on the confluent calculations below.

□

### Optional confluent structure

The origin and threshold calculations give independent information about the first order-five failure, but are not used in the exact-order assembly. Put

$$\mathsf{H}_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \qquad s_j = \Phi^{(j)}(0).$$

Since $\Phi$ is even, simultaneous parity permutations of the rows and columns give

$$\mathsf{H}_4 = AB, \qquad \mathsf{H}_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \qquad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

**Lemma 11.2.** *For the Riemann kernel,*

$$445 < A < 446, \qquad 189223 < B < 189228, \qquad -674000000 < C < -614000000.$$

*In particular,* $\mathsf{H}_4 > 0$ *and* $\mathsf{H}_5 < 0$*.*

*Proof.* For $x = \pi n^2 e^{2u}$, the $n$-th summand of $\Phi(u)$ is $e^{u/2-x}T_0(x)$, where $T_0(x) = 2x(2x-3)$. If its $j$-th derivative is $e^{u/2-x}T_j(x)$, then

$$T_{j+1} = 2xT_j' + (1/2 - 2x)T_j.$$

Thus

$$s_j = \sum_{n\ge 1} e^{-\pi n^2} T_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ supplies rational bounds for $\pi$, and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let $S_j$ be the sum of the absolute coefficients of $T_j$. Since $\deg T_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^{8} 3^r/r! > 20$,

$$\sum_{n\ge 4} |T_j(\pi n^2)| e^{-\pi n^2} \le \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \qquad (j = 0, 2, 4, 6, 8).$$

All endpoints lie on $10^{-36}\mathbb{Z}$, and every arithmetic operation is rounded outward. Substitution into the displayed formulas gives the entirely rational sign chain

$$0 < 445 < A < 446, \qquad 0 < 189223 < B < 189228,$$

$$-674000000 < C < -614000000 < 0.$$

Thus $AB > 0$ and $CB < 0$ without a floating-point determinant. The full grid endpoints and the standard-library verifier are included in the reproducibility archive. $\square$

Applying Lemma 6.2 with $k = 5$, $z = 0$, and $a_i = b_i = i$ shows independently that $\det[\Phi((i-j)\varepsilon)]_{i,j=0}^4 < 0$ for every sufficiently small positive $\varepsilon$. This implication is not needed for Theorem 11.1, whose witness is the fixed rational spacing above.

For the equal-step family put

$$D_5(u,h) = \det[\Phi(u + (i-j)h)]_{i,j=0}^4, \qquad C_5(u) = \lim_{h\downarrow 0} \frac{D_5(u,h)}{h^{20}}.$$

Because $V(0,\ldots,4) = \prod_{j=1}^4 j!$, the Vandermonde and factorial factors in Lemma 6.2 cancel, and therefore

$$C_5(u) = \det[\Phi^{(i+j)}(u)]_{i,j=0}^4. \tag{11.2}$$

**Theorem 11.3** (Confluent order-five threshold)**.** *There is a unique $u_* > 0$ such that $C_5(u_*) = 0$, and*

$$0.0622795266356 < u_* < 0.0622795266357.$$

*Moreover $C_5(u) < 0$ for $0 \le u < u_*$, while $C_5(u) > 0$ for $u > u_*$.*

*Proof.* Put $x = \pi e^{2u}$, $y = e^{-3x}$, and $z = e^{-8x}$. After removing the positive first-mode amplitude $2xe^{5u/2-x}$, the first three theta modes of the derivative jet are

$$b_j = P_j(x) + 4yP_j(4x) + 9zP_j(9x),$$

where

$$P_0(x) = 2x - 3, \qquad P_{j+1}(x) = \left(\frac{5}{2} - 2x\right) P_j(x) + 2xP_j'(x).$$

Thus the three-mode part of $C_5$, apart from a positive fifth power, is the closed determinant

$$G_3(x, y, z) = \det[b_{i+j}]_{i,j=0}^{4}. \tag{11.3}$$

Its exact expansion has 231 monomials. The path derivative is obtained without another expansion from

$$\frac{d}{du} = 2x(\partial_x - 3y\partial_y - 8z\partial_z).$$

For $j \le 9$, the omitted modes satisfy

$$|e_j(x)| \le \frac{S_j x^{j+1} 4^{2j+4} e^{-15x}}{1 - (5/4)^{22} e^{-9x}}, \tag{11.4}$$

where $S_j$ is the sum of the absolute coefficients of $P_j$. Exact rational Bernstein bounds, preserving the common $x$-dependence of $e^{-3x}$ and $e^{-8x}$, give $C_5 < 0$ from $u = 0$ through $u = 1/50$ and $C_5' > 0$ on the overlapping range from $u = 3/200$ through $x = 4$. Direct four-mode endpoint enclosures give opposite signs at $u = 0.0622795266356$ and $u = 0.0622795266357$. Four further Bernstein cells give $C_5 > 0$ for $4 \le x \le 40$, and the exact one-mode determinant plus (11.4) gives positivity for $x \ge 40$. These overlapping signs prove existence, uniqueness, and the stated classification. The domains and cell counts are listed in Appendix F. □

The outermost center over all finite spacings is a separate two-variable extremal problem; no such global extremality is used here.

Equation (11.1) proves part (2) of Theorem 1.1. Together with the strict minors through order four, this completes the exact-order classification. The preceding confluent theorem is not used in that assembly.

*Remark* 11.4 (Riemann Hypothesis boundary)*.* Schoenberg's transform theorem says that the bilateral Laplace transform of a $\mathrm{PF}_\infty$ function is the reciprocal of a Laguerre–Pólya entire function on its strip of convergence [1, 3]. The Fourier transform of $\Phi$, up to the standard normalization, is the Riemann $\Xi$-function, which has known real zeros; hence $\Phi$ is unconditionally not $\mathrm{PF}_\infty$. The Schoenberg formulation related to the Riemann Hypothesis instead asks for the inverse transform of the reciprocal of $\Xi$ to be a Pólya-frequency function [8]. Thus the finite-order classification proved here is a different statement. The next section further shows that strict $\mathrm{PF}_4$ alone does not imply Fourier real-rootedness.

## 12 An explicit continuous $\mathrm{PF}_4$ separator

We now prove Theorem 1.2. Put

$$c=\frac{1}{64},\qquad B(w)=\sum_{k=-3}^{3}b_kw^k,\qquad (b_{-3},\dots,b_3)=(2,10,23,30,23,10,2),$$

and define

$$f_*(x)=e^{-cx^2}B(e^{2cx})=\sum_{k=-3}^{3}b_ke^{ck^2}e^{-c(x-k)^2}. \tag{12.1}$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of $(b_k)$ proves evenness.

### 12.1 Strict $\mathrm{PF}_3$

For fixed $x$, give $\mathsf{X}\in\{-3,\dots,3\}$ probability proportional to $b_ke^{2ckx}$, and write $\chi_j$ for its cumulants. Differentiation of the finite log-partition function gives

$$q=2c-(2c)^2\chi_2,\qquad q'=-(2c)^3\chi_3,\qquad q''=-(2c)^4\chi_4. \tag{12.2}$$

If $\bar{x}=\mathbb{E}\mathsf{X}$, then $0\le\mathbb{E}[(\mathsf{X}+3)(3-\mathsf{X})]=9-\chi_2-\bar{x}^2$. Thus $0\le\chi_2\le 9$, and hence

$$q\ge 2c(1-18c)=2c\frac{23}{32}>0. \tag{12.3}$$

Moreover,

$$\begin{aligned}F_2&=q^3-\{qq''-(q')^2\}\\&=q^3+q(2c)^4\chi_4+(2c)^6\chi_3^2.\end{aligned} \tag{12.4}$$

The identity $\chi_4=\mathbb{E}(\mathsf{X}-\mathbb{E}\mathsf{X})^4-3\chi_2^2$ gives $\chi_4\ge -243$. Dropping the final nonnegative term in (12.4) gives

$$F_2\ge q\{q^2-3888c^4\}.$$

At $c=1/64$, the two coefficients after division by $c^2$ are

$$q^2/c^2\ge\frac{529}{256},\qquad 3888c^2=\frac{243}{256}.$$

Thus $F_2\ge qc^2(143/128)>0$ globally. The weighted-mean criterion of Section 5 proves strict $\mathrm{PF}_3$.

### 12.2 Exact order-four density

Let $\epsilon=2c=1/32$, $t=e^{\epsilon x}$, and

$$P(t)=t^3B(t)=2+10t+23t^2+30t^3+23t^4+10t^5+2t^6.$$

For $E=t\frac{\mathrm{d}}{\mathrm{d}t}$, define

$$N_1=tP',\qquad N_{j+1}=t(N_j'P-jN_jP'). \tag{12.5}$$

The quotient rule gives $E^j\log P=N_j/P^j$. Therefore, for $\ell_j=(\log f_*)^{(j)}$,

$$\ell_2=\frac{-\epsilon P^2+\epsilon^2N_2}{P^2},\qquad \ell_j=\frac{\epsilon^jN_j}{P^j}\quad(3\le j\le 6). \tag{12.6}$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so every term may first be put over the common denominator $P^{12}$. The resulting numerator has the exact factor $P^8$. After cancellation,

$$\mathcal{C}_4[f_*](x) = \frac{\mathcal{G}(t)}{2^{54} P(t)^4}. \tag{12.7}$$

The exact expansion printed in Appendix E proves that $\mathcal{G}$ is palindromic of degree 24 and that all 25 integer coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

The established inequalities $q > 0$ and $F_2 > 0$ also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for $f_*$. Thus $\partial_\xi \Psi < 0$, and Proposition 6.3 proves that $f_*$ is strict $\mathrm{PF}_4$.

### 12.3 Nonreal Fourier zeros

With $\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-izx}\,\mathrm{d}x$, Gaussian translation gives

$$\widehat{f_*}(z) = \sqrt{\frac{\pi}{c}} e^{-z^2/(4c)} \sum_{k=-3}^{3} b_k e^{ck^2} e^{-ikz}. \tag{12.8}$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \tag{12.9}$$

vanishes. At $c = 1/64$, the exact calculation in Appendix E gives $\operatorname{disc}(R_c) < 0$. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w + w^{-1}$ is real. The corresponding $w$-roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\operatorname{Im} z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.2.

## 13 Exact verification and reproduction

The primary universal theorem is checked by the pinned Lean project in `proof/formal`. From that directory, `lake build` constructs the real theta series, analytic kernel and global derivative jet, universal cleared signs, fixed-order quotient and transport chain, arbitrary-node strict minors through order four, direct negative order-five witness, and exact-order assembly. The focused command `lake env lean PF4/Audit.lean` prints the axiom dependencies of the exported declarations. They are exactly `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, custom axiom, or unsafe theorem bridge occurs in the target dependency tree.

The principal claim boundaries are:

| Object | Maintained boundary |
|---|---|
| Analytic kernel and global jet | PF4.globalRiemannKernel_eq_thetaSeries_abs; PF4.GlobalKernelJetIdentification.iteratedDeriv_ globalRiemannKernel_eq_thetaSeriesJet |
| Global $q, F_2, \mathcal{C}_4$ | PF4.CERT12OuterClosure.normalized_cleared_signs_pos and exact raw-jet transfer |
| Lower $\Lambda$ and orders two–three | PF4.WeightedMeanVariation.actual_weightedMeanVariation_ lowerLambda; PF4.GlobalStrictPF4.translationMinor_two_pos; PF4.GlobalStrictPF4.translationMinor_three_pos |
| Fixed-order quotient cascade | PF4.FixedOrderQuotientWronskian.fixedOrderFour_ quotientWronskian_package |
| Actual coordinate transport | PF4.GlobalStrictPF4.globalRiemannKernel_ coordinateNumerator_pos; PF4.GlobalStrictPF4. globalRiemannKernel_coordinatePartialXiPsi_neg |
| Universal strict PF4 | PF4.globalRiemannKernel_strictPFUpTo_four |
| Direct PF5 witness | PF4.globalRiemannKernel_orderFive_translationMinor_neg |
| Exact order four | PF4.globalRiemannKernel_pfOrderExactly_four |

Independent exact verifiers reconstruct the printed finite algebra and rational inequalities:

| Object | Repository replay | Role |
|---|---|---|
| Global $q, F_2, \mathcal{C}_4$ | scripts/verify_riemann_signs_exact.py | exact arithmetic |
| Fixed Wronskian algebra | scripts/verify_pf4_wronskian_reduction.py | independent |
| Endpoint transport algebra | scripts/verify_pf4_transport_kernel.py | independent |
| Direct PF5 witness and optional threshold | scripts/verify_pf5_threshold.py | exact arithmetic |
| Optional threshold polynomial | scripts/verify_pf5_threshold_symbolic.py | independent |
| Continuous separator | scripts/verify_continuous_pf4_separator.py | exact arithmetic |

The global-sign replay retains the first two theta modes exactly. It checks the 361 Bernstein and decay inequalities summarized in Appendix B, the 140 derivative-envelope coefficients, and analytic geometric bounds for every mode $n \geq 3$. It uses a pure-Python rational domain and refuses a FLINT import. No floating-point comparison or interval subdivision decides a sign.

The PF5 replay directly encloses the five kernel values at spacings $0, h, 2h, 3h, 4h$, $h = 211/2000$, and proves the determinant interval (11.1). Its additional cells reconstruct the optional 231-monomial confluent determinant and unique threshold. The separator replay checks the exact $P^8$ cancellation, all 25 coefficients in (12.7), and the rational discriminant inequality in Appendix E.

From the repository root, the independent artifact command is

```
python scripts/replay_paper.py.
```

It prints one `PASS` line per active exact certificate followed by `status=paper evidence replay passed`. Package versions, the complete stdout digest, PDF digest, and arXiv archive digest are recorded in `paper/RELEASE_MANIFEST.md`. Earlier computer-algebra experiments are discovery provenance only; the maintained Lean declarations and exact replay boundaries above carry the final claims.

## 14 Declarations, availability, and computational provenance

**Author contribution.** Joshuah Rainstar conceived and directed the project, developed the proof architecture, verified the exact computations, and prepared the manuscript.

**Acknowledgments and computational assistance.** The author acknowledges the use of OpenAI Codex and ChatGPT, Anthropic Claude, and Moonshot AI Kimi for symbolic calculation, implementation, drafting, and review. The author takes responsibility for the mathematical claims and the final manuscript.

**Conflict of interest.** The author declares no competing interests.

**Code and data availability.** The public repository is `https://github.com/falseywinchnet/zeta`. The submitted manuscript is preserved by the immutable tag `pf4-paper-v1.0.0`. Its commit is

`4a14988093d7afaf05f453f8daf5441c09ba58b3`.

The present revised proof is release `PF4-paper-v2.0.0`, recorded by MIND epoch `P000163`. Their distinct status and material proof changes are listed in `paper/REVISION_HISTORY.md`; current source, environment, replay, PDF, and archive digests are listed in `paper/RELEASE_MANIFEST.md`. The repository archive is assigned DOI `https://doi.org/10.5281/zenodo.21404360`.

The formal replay is `lake build` followed by `lake env lean PF4/Audit.lean` from `proof/formal`. The independent exact arithmetic replay is `python scripts/replay_paper.py` from the repository root.

All theorem-bearing sign decisions are exact. Numerical values produced by the repository scripts are diagnostic only; there is no separate experimental dataset.

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## A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \tag{A.1}$$
$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \tag{A.2}$$

In the curvature coordinate, repeated application of $Q\,\mathrm{d}/\,\mathrm{d}y$ to $c_2 = -Q$ gives

$$\begin{aligned}
c_2 &= -Q,\\
c_3 &= -QQ',\\
c_4 &= -Q\{(Q')^2 + QQ''\},\\
c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\},\\
c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}.
\end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned}
\mathcal{P} = {}& QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'''\\
&+ 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12.
\end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$\{Q(\log\kappa)'\}' = \frac{Q'\kappa' + Q\kappa''}{\kappa} - \frac{Q(\kappa')^2}{\kappa^2},$$

so, after multiplication by $\kappa^2$ and substitution of $\kappa' = -Q'''$, $\kappa'' = -Q''''$,

$$3(\kappa - 1) - \{Q(\log\kappa)'\}' = -\frac{\mathcal{P}}{(Q''-2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2 D$, proving (8.2) by an exposed common numerator.

# B Exact proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has $j$-th derivative

$$2\pi n^2 e^{5u/2-n^2x} R_j(n^2x), \qquad x = \pi e^{2u},$$

where the polynomials $R_j$ satisfy the recurrence in Section 4. Machin's identity and alternating series give $\pi > 157/50$; hence $R_0(x) = 2x - 3 > 0$. The same recurrence, exponential bounds, compact Bernstein margins, and infinite-tail reduction are encoded in the maintained Lean modules `PF4.CERT12CompactClosure` and `PF4.CERT12OuterClosure`. The formulas below remain the human-readable certificate and the independent exact-Python replay.

## B.1 Cleared sign polynomials

In addition to $a_j$, define the normalized remaining modes

$$e_j(x) = \sum_{n\geq 3} \frac{n^2 e^{-(n^2-1)x} R_j(n^2x)}{R_0(x)}, \qquad \widehat{a}_j = a_j + e_j.$$

For an indeterminate raw jet $(\widehat{a}_0, \ldots, \widehat{a}_6)$, put

$$N_q = \widehat{a}_1^2 - \widehat{a}_0\widehat{a}_2,$$

$$\begin{aligned}N_{F_2} = &-\widehat{a}_0^4\widehat{a}_2\widehat{a}_4+\widehat{a}_0^4\widehat{a}_3^2+\widehat{a}_0^3\widehat{a}_1^2\widehat{a}_4-2\widehat{a}_0^3\widehat{a}_1\widehat{a}_2\widehat{a}_3+2\widehat{a}_0^3\widehat{a}_2^3\\ &-3\widehat{a}_0^2\widehat{a}_1^2\widehat{a}_2^2+3\widehat{a}_0\widehat{a}_1^4\widehat{a}_2-\widehat{a}_1^6,\end{aligned}$$

and

$$N_{\mathcal{C}_4}=\det[\widehat{a}_{i+j}]_{i,j=0}^3.$$

Direct logarithmic differentiation and the central-moment determinant give

$$q=\frac{N_q}{\widehat{a}_0^2},\qquad F_2=\frac{N_{F_2}}{\widehat{a}_0^6},\qquad \mathcal{C}_4=\frac{N_{\mathcal{C}_4}}{\widehat{a}_0^4}. \tag{B.1}$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$q_1=\frac{4x(4x^2-12x+15)}{(2x-3)^2},$$

$$(F_2)_1=\frac{64x^3(64x^6-576x^5+2448x^4-6432x^3+10044x^2-8100x+2295)}{(2x-3)^6},$$

$$(\mathcal{C}_4)_1=\frac{49152x^6(16x^4-96x^3+360x^2-840x+945)}{(2x-3)^4}.$$

After $x=5+z$, every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

## B.2 Exact two-mode margin

Rational Taylor bounds for the exponential imply

$$0<\eta<\frac{1}{12000}\quad (x\ge\pi),$$

and

$$\frac{1}{23000}<\eta<\frac{1}{12000}\quad \left(\frac{157}{50}\le x\le\frac{10}{3}\right).$$

The close endpoints are entirely rational checks. Namely,

$$\sum_{k=0}^{16}\frac{(471/50)^k}{k!}>12000,\qquad \sum_{k=0}^{21}\frac{10^k}{k!}>22000,$$

and the geometric bound for the tail after degree 13 gives

$$e^{10}<\sum_{k=0}^{13}\frac{10^k}{k!}\frac{10^{14}/14!}{1-10/15}=\frac{3480060751}{154791}<23000.$$

These prove, respectively, the upper $\eta$-cap, the cap $e^{-10}<1/22000$, and the lower bound $e^{-10}>1/23000$. Likewise $\sum_{k=0}^{21}15^k/k!>3{,}000{,}000$, which gives $\eta<1/3{,}000{,}000$ for $x\ge 5$. We use the following elementary certificate lemma. If $p(s,t)=\sum p_{ij}s^it^j$ has bidegree $(d,e)$, its Bernstein coefficients on the unit square are

$$\beta_{k\ell}=\sum_{i\le k,\,j\le \ell}p_{ij}\frac{\binom{k}{i}}{\binom{d}{i}}\frac{\binom{\ell}{j}}{\binom{e}{j}}.$$

The power-to-Bernstein identity proves that $\beta_{k\ell} > 0$ for every $k, \ell$ implies $p > 0$ on the square.

Apply this lemma after affine rational changes of variables. For $q$, one rectangle ends at $x = 10/3$, followed by a positive base polynomial minus exponentially decreasing corrections. For $F_2$, dependent bounds on the first rectangle are followed by a rectangle through $x = 5$ and a positive half-strip expansion. For $\mathcal{C}_4$, the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000. \tag{B.2}$$

This is a finite algebraic proof on fixed domains.

Here is the precise connection between those coefficient checks and the quantitative margins. Evaluate the raw polynomials at the two-mode jet and set

$$\mathcal{Q} = N_q - 10a_0^2, \qquad \mathcal{F} = N_{F_2} - 1000a_0^6, \qquad \mathcal{C} = N_{\mathcal{C}_4} - 50{,}000{,}000a_0^4.$$

The rows labelled $Q, F, C$ below certify the cleared numerators of $\mathcal{Q}, \mathcal{F}, \mathcal{C}$, respectively. Thus their positivity is exactly, rather than merely qualitatively, the three inequalities in (B.2). For a ratio row, write the shifted target as $B_0(x) + \sum_{j\geq 1} \eta^j B_j(x)$, where $B_0$ has positive shifted coefficients. Whenever $B_j$ is adverse, put $D_j = -B_j$. Positivity of

$$3jD_jB_0 - D_j'B_0 + D_jB_0'$$

after the same shift proves that $e^{-3jx}D_j(x)/B_0(x)$ decreases. The displayed endpoint ratio is the sum of all such adverse ratios at the left endpoint; being less than one proves the target positive on the entire half-line. Coefficientwise nonnegative pieces only increase it.

For reference, the complete finite certificate is summarized below. Write $\mathcal{A} = (2x-3)+4(8x-3)\eta$, which is positive on every listed domain. Rows $Q_0, F_0, F_1, C_0$ are Bernstein certificates after the indicated affine change to the unit square; the remaining rows use coefficient positivity after a half-line shift. In the two "ratio" rows, the listed fraction is the sum of the adverse coefficient ratios and is strictly less than one.

| ID | Variables and domain | Degree | Positive denominator | Exact lower datum |
|---|---|---|---|---|
| $Q_0$ | $157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$ | $(4, 2)$ | $\mathcal{A}^2$ | 6613625187767/70312500000 |
| $Q_\infty$ | $x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$ | $(4, 2)$ | $\mathcal{A}^2$ | ratio 47333/505500 |
| $F_0$ | $157/50 \leq x \leq 10/3,\ 1/23000 \leq \eta \leq 1/12000$ | $(12, 6)$ | $\mathcal{A}^6$ | minimum > 1610047 |
| $F_1$ | $10/3 \leq x \leq 5,\ 0 \leq \eta \leq 1/22000$ | $(12, 6)$ | $\mathcal{A}^6$ | 29124973000/19683 |
| $F_\infty$ | $x = 5 + z,\ \eta = v/3000000,\ 0 \leq v \leq 1$ | $(12, 6)$ | $\mathcal{A}^6$ | 432/19073486328125 |
| $C_0$ | $157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$ | $(14, 4)$ | $\mathcal{A}^4$ | minimum > 9546672947 |
| $C_\infty$ | $x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$ | $(14, 4)$ | $\mathcal{A}^4$ | ratio 65989337396066791/77165720550000000 |

These seven checks comprise 361 exact rational inequalities. The derivative envelopes used below contribute 140 further Bernstein coefficients, of bidegrees $(j+1, 1)$, $0 \leq j \leq 6$; their smallest coefficient is 44861/31250. Thus the compact domains and the differentiated tail majorants belong to the same exact certificate.

### B.3 All later modes

For direct reconstruction of the derivative bounds, put $S_j(t) = 2^j(-1)^j R_j(t)$. The recurrence gives

$$
\begin{aligned}
S_0 &= 2t - 3,\\
S_1 &= 8t^2 - 30t + 15,\\
S_2 &= 32t^3 - 224t^2 + 330t - 75,\\
S_3 &= 128t^4 - 1440t^3 + 4232t^2 - 3270t + 375,\\
S_4 &= 512t^5 - 8448t^4 + 41408t^3 - 68096t^2 + 30930t - 1875,\\
S_5 &= 2048t^6 - 46592t^5 + 343040t^4 - 976320t^3 + 1008968t^2 - 285870t + 9375,\\
S_6 &= 8192t^7 - 245760t^6 + 2536960t^5 - 11109120t^4 + 20633312t^3 - 14260064t^2 + 2610330t - 46875.
\end{aligned}
$$

For $t \geq 27$, shifting $t = 27 + z$ gives positive coefficients in $(-1)^j R_j(t)$ and in $2^{j+1}t^{j+1} - (-1)^j R_j(t)$. Therefore

$$0 < (-1)^j R_j(t) \leq 2^{j+1}t^{j+1}, \qquad 0 \leq j \leq 6. \tag{B.3}$$

The derivative numerator of the normalized $n = 3$ term is another positive shifted polynomial, so its absolute value decreases for $x \geq 3$. For $n \geq 4$, successive majorants have ratio at most

$$\left(\frac{5}{4}\right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for $e^{24}, e^{27}, e^{45}$ then yield, on $\pi \leq x \leq 5$,

$$(|e_0|, \ldots, |e_6|) < (7\cdot 10^{-9}, 4\cdot 10^{-7}, 2\cdot 10^{-4}, 7\cdot 10^{-4}, 3\cdot 10^{-2}, 1.1, 41). \tag{B.4}$$

Indeed, the normalized $n = 3$ term decreases on $x \geq 3$, and $e^{24} > 25{,}000{,}000{,}000$, so it is bounded by $3|R_j(27)|/25{,}000{,}000{,}000$. The sum of the $n \geq 4$ majorants is at most

$$\frac{2^{j+2}3^j 4^{2j+4}}{3 \cdot 10^{19}}.$$

The data entering these two expressions are

| $j$ | $\lvert R_j(27)\rvert$ | chosen upper bound for $\lvert e_j\rvert$ |
|---|---|---|
| 0 | 51 | $7 \cdot 10^{-9}$ |
| 1 | 5037/2 | $4 \cdot 10^{-7}$ |
| 2 | 475395/4 | $2 \cdot 10^{-4}$ |
| 3 | 42678141/8 | $7 \cdot 10^{-4}$ |
| 4 | 3623251731/16 | $3 \cdot 10^{-2}$ |
| 5 | 288709329165/32 | 11/10 |
| 6 | 21373315648611/64 | 41 |

For each row the later-mode expression is less than one thousandth of the $n = 3$ expression, and their sum is strictly below the chosen bound. This derives every constant in (B.4) from the printed recurrence and three rational exponential inequalities. The same Bernstein lemma gives

$$(|a_0|, \ldots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

For any one of the printed raw polynomials $N = \sum_\alpha c_\alpha X^\alpha$, the perturbation estimate used here is the explicit finite sum

$$|N(a+e) - N(a)| \le \sum_\alpha |c_\alpha| \left\{ \prod_{j=0}^{6} (A_j + E_j)^{\alpha_j} - \prod_{j=0}^{6} A_j^{\alpha_j} \right\}, \tag{B.5}$$

whenever $|a_j| \le A_j$ and $|e_j| \le E_j$. Substitution of the two displayed seven-tuples into (B.5) gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{\mathcal{C}_4}| < 31{,}549{,}532. \tag{B.6}$$

For $x \ge 5$, (B.3) gives

$$|a_j| \le 2^{j+2} x^j, \qquad |e_j| \le 2^{j+2} 3^{2j+4} x^j e^{-8x}.$$

The raw weights of $N_q, N_{F_2}, N_{\mathcal{C}_4}$ are respectively $2, 6, 12$. Thus every perturbation majorant is a positive multiple of $x^w e^{-8x}$, or a faster-decaying term, with $w \le 12$; it decreases for $x \ge 5$. In (B.5) take

$$A_j = 2^{j+2} 5^j, \qquad E_j = \frac{2^{j+1} 3^{2j+4} 5^j}{10^{17}}.$$

The latter follows from the displayed tail bound and $\sum_{k=0}^{47} 40^k/k! > 2 \cdot 10^{17}$. Direct substitution gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{\mathcal{C}_4}| < 418{,}287. \tag{B.7}$$

To compare like quantities, note that

$$a_0 = 1 + 4\eta \frac{R_0(4x)}{R_0(x)} > 1,$$

and, by homogeneity,

$$N_{q,1,2} = q_{1,2} a_0^2, \qquad N_{F_2,1,2} = (F_2)_{1,2} a_0^6, \qquad N_{\mathcal{C}_4,1,2} = (\mathcal{C}_4)_{1,2} a_0^4.$$

Consequently (B.2) supplies the same respective lower margins 10, 1000, and 50,000,000 for the cleared numerators. Both perturbation bounds (B.6) and (B.7) are strictly smaller. Their scale relative to the corresponding cleared-numerator margin is recorded here:

| target | surviving fraction on $\pi \le x \le 5$ | surviving fraction on $x \ge 5$ |
|---|---|---|
| $q$ | $> 0.9999595$ | $> 0.99999999999351$ |
| $F_2$ | $> 0.962041$ | $> 0.99997$ |
| $\mathcal{C}_4$ | $> 0.36900936$ | $> 0.99163426$ |

Thus even the least relative margin, the compact $\mathcal{C}_4$ bound, retains more than 36.9 percent of its certified two-mode numerator margin. Finally $\widehat{a}_0 = a_0 + e_0 > 0$, so (B.1) proves all three signs for $u \ge 0$. Evenness proves Lemma 4.1.

# C Endpoint algebra for the transport cancellation

This appendix records the algebra suppressed by the phrase "simplifying the endpoint expression." Put

$$z = p + L, \qquad w = z + R, \qquad a = \frac{Q(z) - Q(p)}{L}, \qquad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \qquad \Lambda = L + R + a - b. \tag{C.1}$$

Define

$$\mathcal{J}_y = y^3 - 3yQ(y) + Q(y)Q'(y),$$
$$\mathcal{H}_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}.$$

With $I_L = \mathcal{J}_z - \mathcal{J}_p$ and $I_R = \mathcal{J}_w - \mathcal{J}_z$, the identity needed in Lemma 9.1 is exactly

$$\begin{aligned}&\frac{-(\mathcal{J}_z - \mathcal{J}_p)/L + (\mathcal{J}_w - \mathcal{J}_z)/R}{\Lambda} + \frac{L\mathcal{H}_p - (\mathcal{J}_z - \mathcal{J}_p)}{L^2\delta} \\ &\quad = \Lambda + Q'(p) - a + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)(2 - Q''(p))}{L\delta}.\end{aligned}$$

To check it, substitute (C.1), replace $z$ by $p+L$ and $w$ by $p+L+R$, and multiply by the common denominator $L^2R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for $Q$ is used, so the identity holds for arbitrary smooth $Q$. Equivalently, regard the endpoint jets $Q(p), Q(z), Q(w), Q'(p), Q'(z), Q'(w), Q''(p)$ and $p, L, R$ as independent indeterminates subject only to the displayed definitions of $a, b, \delta$, and $\Lambda$. Clearing $L^2R\delta\Lambda$ first and then expanding gives the zero polynomial. Expanding the uncleared rational expression need not expose this cancellation.

# D Claim provenance

Every theorem-sized claim in the main proof has one of three boundaries: a maintained Lean declaration, an exact local certificate, or a standard argument printed in full. The following table is the audit trace.

| Claim | Boundary | Exact anchor |
|---|---|---|
| Theta, Poisson, analyticity, parity, global jet | Lean/mathlib | `PF4.KernelAnalytic`; `PF4.GlobalKernelJetIdentification`; `CERT20--CERT22` |
| Global $q, F_2, \mathcal{C}_4$ | Lean plus exact rational replay | `PF4.CERT12OuterClosure.normalized_cleared_signs_pos`; `CERT22` |
| Weighted-mean lower $\Lambda$ | Lean | `PF4.WeightedMeanVariation.actual_weightedMeanVariation_lowerLambda`; `CERT28` |
| Fixed $W_3, W_4$ identities | Lean and direct algebra | `PF4.FixedOrderQuotientWronskian`; `CERT29` |
| Sizes two–four quotient integrals | Lean plus printed FTC proof | `PF4.GenericQuotientIntegral`; `PF4.ContinuousQuotientBox` |
| Confluent limits | standard printed proof | Lemma 6.2; successive divided differences |
| Actual coordinate and translation | Lean | `PF4.CurvatureCoordinateRealization`; `PF4.SimultaneousCoordinateTranslation`; `CERT19`, `CERT27` |
| $\mathcal{C}_4 = Q^6\kappa^2 D$ | Lean and printed algebra | `PF4.C4Invariant`; Appendix A; `CERT25--CERT26` |
| Endpoint cancellation and closed gap | Lean and printed integration | `PF4.LocalGapClosure`; `PF4.LocalCentralIntegration`; `CERT19`, `CERT25` |
| Strict PF4 at arbitrary nodes | Lean | `PF4.globalRiemannKernel_strictPFUpTo_four`; `CERT23` |
| Direct negative PF5 minor | Lean plus exact rational replay | `PF4.globalRiemannKernel_orderFive_translationMinor_neg`; `CERT24` |
| Exact order four | Lean | `PF4.globalRiemannKernel_pfOrderExactly_four`; `CERT24` |
| Optional PF5 threshold | exact local certificate | `CERT16--CERT17`; Appendix F |
| Continuous Fourier separator | exact local certificate | `CERT10`; Appendix E |

The detailed declaration-by-declaration map is `proof/CLAIM_INDEX.md`; the target axiom audit is `proof/formal/AXIOMS.md`. The frozen submitted version and this revised proof are identified separately in `paper/REVISION_HISTORY.md`.

# E Exact separator calculation

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants $\ell_2, \dots, \ell_6$, the verifier reconstructs the central moments

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = \ell_2, \quad m_3 = \ell_3,$$

$$m_4 = \ell_4 + 3\ell_2^2, \qquad m_5 = \ell_5 + 10\ell_3\ell_2,$$

$$m_6 = \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over $\mathbb{Q}[t]$. Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator $P^{12}$. After collecting, the numerator is divisible by $P^8$ in $\mathbb{Q}[t]$. Cancel this factor and clear the remaining powers of two, writing

$$\mathcal{C}_4[f_*](x) = \frac{\mathcal{G}(t)}{2^{54}P(t)^4}.$$

Exact rational arithmetic verifies

$$\deg \mathcal{G} = 24, \qquad [t^j]\mathcal{G} = [t^{24-j}]\mathcal{G} > 0 \quad (0 \le j \le 24),$$

with smallest integer coefficient

$$\min_{0\le j\le 24}[t^j]\mathcal{G} = 3221225472.$$

Equivalently, the smallest coefficient before multiplication by $2^{54}$ is $3/2^{24}$. Write

$$\mathcal{G}(t) = \sum_{j=0}^{24} a_j t^j, \qquad a_j = a_{24-j}.$$

The 13 independent integer coefficients are:

| $j$ | $a_j$ | $j$ | $a_j$ |
|---|---:|---|---:|
| 0 | 3221225472 | 7 | 357704506872240 |
| 1 | 61435985920 | 8 | 701640591264080 |
| 2 | 575070935296 | 9 | 1172880923200640 |
| 3 | 3512452441920 | 10 | 1684288128586089 |
| 4 | 15702254784944 | 11 | 2088645133531440 |
| 5 | 54633824917920 | 12 | 2243243092164284 |
| 6 | 153607464071178 | | |

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `d0f91259919c7f23`; the complete hash is recorded in the release manifest. Coefficient positivity proves the sign on the full domain $t > 0$. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with $\mathcal{G}/(2^{54}P^4)$.

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. Put

$$r = e^{1/64}, \qquad x = r^2 = e^{1/32}.$$

Exact expansion gives

$$\operatorname{disc}(R_c) = 4r^{10}G(x),$$

where

$$\begin{aligned} G(x) = {} & 432x^{13} - 4968x^9 + 900x^8 + 16200x^6 + 19044x^5 \\ & - 72600x^4 + 20000x^3 + 62100x^2 - 54334x + 13225. \end{aligned}$$

Writing $y = x - 1$ gives

$$\begin{aligned} G(1+y) = {} & -1 - 10y - 12y^2 + 680y^3 + 11532y^4 + 96660y^5 \\ & + 365400y^6 + 569664y^7 + 512172y^8 + 303912y^9 \\ & + 123552y^{10} + 33696y^{11} + 5616y^{12} + 432y^{13}. \end{aligned}$$

Moreover $0 < y < 1/30$. Indeed,

$$e < 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18} < \frac{11}{4},$$

where $1/18 = (1/24)\sum_{j\ge 0} 4^{-j}$ bounds the tail beginning with $1/4!$. The binomial theorem gives

$$\frac{11}{4} < 1 + \frac{32}{30} + \frac{\binom{32}{2}}{30^2} + \frac{\binom{32}{3}}{30^3} = \frac{1891}{675} < \left(\frac{31}{30}\right)^{32}.$$

The sum of the positive terms at $y = 1/30$ is exactly

$$\sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} = \frac{1621193195302591}{3690562500000000} < 1.$$

The two omitted terms $-10y - 12y^2$ are negative. Consequently

$$G(1+y) < -1 + \sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} < 0,$$

and hence $\operatorname{disc}(R_c) < 0$ without numerical evaluation.

# F Direct witness and optional order-five threshold

The direct finite witness in (11.1) is the order-five input to Theorem 1.1. The remaining cells in this appendix certify the optional Theorem 11.3. All endpoints lie on $10^{-50}\mathbb{Z}$, and every product and rational scaling is rounded outward on that lattice. Machin's identity bounds $\pi$; positive Taylor sums with a geometric remainder bound $e^{2u}$; and argument division with alternating sums bounds the negative exponentials.

The recurrence in the theorem reconstructs $P_0, \dots, P_{10}$ over $\mathbb{Q}[x]$. Independently expanding the determinant (11.3) gives a polynomial of degree 23 in $x$, degree at most five in each of $y, z$, and 231 nonzero monomials. On a cell $x = a + w$, $0 \le w \le d \le 1/10$, the verifier keeps the correlation between the exponential variables by using

$$e^{-rx} = e^{-ra} \sum_{k=0}^{8} \frac{(-rw)^k}{k!} + E_r(w), \qquad |E_r(w)| \le e^{-ra} \frac{(rd)^9}{9!}, \qquad r = 3, 8.$$

It substitutes these interval polynomials into $G_3$ and converts the result to the Bernstein basis in $w$. The later-mode boxes (11.4) are propagated through the 120 determinant monomials.

| Object | Domain | Cells | Certified sign |
|---|---|---|---|
| $C_5$ | $0 \le u \le 1/50$ | 2 | negative |
| $C_5'$ | $3/200 \le u,\ x \le 4$ | 22 | positive |
| root endpoints | $u = 0.0622795266356,\ 0.0622795266357$ | 2 | $(-,+)$ |
| $C_5$ | $4 \le x \le 40$ | 4 | positive |
| one-mode tail | $x \ge 40$ | 1 | positive |
| finite witness | $u = 0,\ h = 211/2000$ | 1 | negative |

At the two root endpoints the fourth theta mode is retained explicitly and only $n \ge 5$ is bounded. This gives the rational sign margins needed for the interval of width $10^{-13}$. For $x \ge 40$, the one-mode determinant is

$$2 \cdot 150994944\, x^{10}(32x^5 - 240x^4 + 1200x^3 - 4200x^2 + 9450x - 10395).$$

After subtracting $x^5$, the polynomial has positive coefficients in $x - 40$. The remaining modes are smaller than this lower bound because $x^{10}e^{-3x}$ decreases there; the verifier checks the resulting exact rational ratio is below one.

The symbolic reconstruction is `scripts/verify_pf5_threshold_symbolic.py`. The independent lattice, Bernstein, tail, root, and finite-witness replay is `scripts/verify_pf5_threshold.py`.